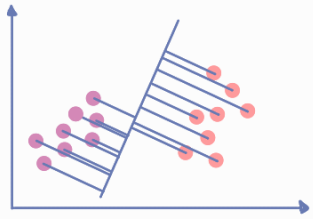
Fisher's linear discriminant (LDA - Linear Discriminant Analysis)



This model does classification in terms of dimensionality reduction. LDA uses the information from all the features to create a new axis and projects the data over it in a way that it maximizes the separation of the classes.

The simplest measure of the separation of classes when projected onto w, is the separation of the projected class means.

$$\text{maximize } J(w) = w^T(m_2 - m_1)$$

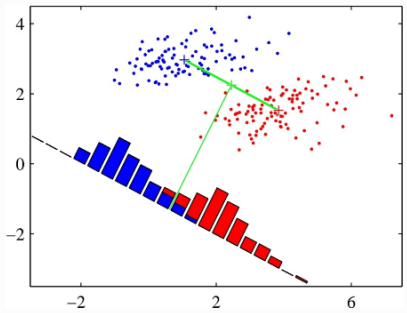
$$m_1 = \frac{1}{N_1} \sum_{n|x_n \in C_1} x_n \qquad m_2 = \frac{1}{N_2} \sum_{n|x_n \in C_2} x_n$$

FOR 2 CLASSES

this expression can be made arbitrarily large simply by increasing the magnitude of w; constrain w to unit length $\rightarrow |w| = 1$

Consider the case in which we have only two classes; the concept is the same for a multiclass problem:

1. compute the mean for the two distributions;
2. plot the line connecting the two means;
3. project all the points onto this line;
4. choose a threshold and classify the points on its left as one class and the points on its right as the other class.

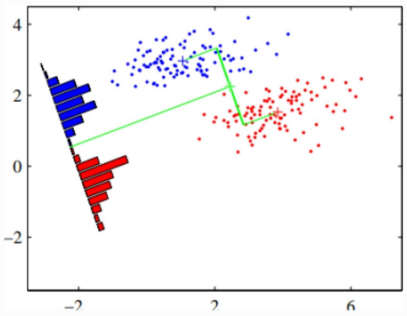


$$w \propto (m_2 - m_1)$$

We use the mean but we are not using the covariance. Add information on the covariance allowing the line also to rotate

$$w \propto S_W^{-1}(m_2 - m_1)$$

Rotation matrix



$$w_0 = w^T m$$

with m = global mean of all the dataset

Once we compute weight matrix and w_0

PREDICTION

Fisher's linear discriminant is given by the function $y = w^T x$ and the classification of new instances is given by $y > -w_0$

PCA $\longrightarrow$ reduces dimensions by focusing on the features with the most variation
LDA $\longrightarrow$ focuses on maximizing the separability among classes

Multiple classes:

$$y = W^T x$$

$$J(W) = Tr\{(W S_W W^T)^{-1}(W S_B W^T)\}$$

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